

HVK1D/HVK2D Q1-Validation Addendum: Held-Out CIFAR Baselines, Observable Ablations, and Shot-Noise Robustness

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Abstract—This addendum strengthens the HVK1D/HVK2D evidence package by separating diagnostic pair-correlation performance from held-out natural-image reconstruction. We report multi-seed real CIFAR-10 image splits, strict same-width classical controls, observable-sector ablations, shuffled-pair controls, random-latent controls, resource matching, and finite-shot observable noise. Unlike the restricted pair-correlation diagnostic, the real held-out CIFAR test does not establish quantum advantage: the best real-image result is obtained by local-observables-only, while HVK2D-real-cifar reaches 19.62 dB mean PSNR. The correct conclusion is therefore narrower: HVK2D has a useful entanglement-sensitive diagnostic, but the current real-image validation still requires architectural improvement before a Q1-level quantum-advantage claim is defensible.

Index Terms—Hamiltonian vision kernel, CIFAR-10 reconstruction, ablation study, quantum observables, shot noise, classical controls.

I. VALIDATION PROTOCOL

Ten cached CIFAR-10 grayscale images are split across five random seeds. For each seed, six images are used to fit a single shared linear readout from a 32-dimensional latent feature vector to 8×8 patch pixels, and four held-out images are reconstructed without per-image retraining. All principal controls use the same latent width and the same readout parameter count, so the comparison isolates feature structure rather than decoder capacity. This cached-image protocol is a reproducible smoke test, not a substitute for the recommended class-balanced CIFAR-10 subset protocol with mutually exclusive training, validation, and test images.

II. CLASSICAL PAIR-FEATURE CONTROLS

Because the nonlocal target is constructed from pair products, a raw-linear baseline is not by itself a sufficient classical control. We therefore include explicit quadratic-feature controls in the CIFAR reconstruction suite and in the CIFAR-derived nonlocal diagnostic. These controls test whether performance arises from access to multiplicative pair features generally, rather than from a uniquely quantum mechanism.

III. HELD-OUT CIFAR RESULTS

IV. STATISTICAL ANALYSIS

All principal image-level comparisons are paired by seed and held-out image. For each pair, we compute the PSNR difference between HVK2D-real-cifar and the corresponding control. Because the sample size is small and normality should

not be assumed, we report a Wilcoxon signed-rank p value and a bootstrap 95% confidence interval for the mean paired PSNR difference. Positive values favor HVK2D; negative values favor the control.

V. SHOT-NOISE ROBUSTNESS

VI. INTERPRETATION

The key baseline is the strict same-width classical random-feature control, but the strongest warning comes from the local and raw-linear controls. In the real held-out CIFAR table, local-observables-only obtains the best mean MSE ($1.0024e - 02$), while HVK2D-real-cifar is weaker ($1.2197e - 02$). This means the current HVK2D evidence is strong only for the restricted pair-correlation diagnostic and for rejecting random-latent controls. It does not yet support a broad natural-image quantum-advantage claim. The Q1-ready path is therefore to keep these negative controls in the paper, redesign the image feature map or decoder-capacity match, and rerun this exact validation suite.

VII. REPRODUCIBILITY

The full tables, plots, resource comparison, and CSV files are generated under `main2/newHVK/results/q1_validation/` by running `main2/newHVK/run_newhvk_suite.py --q1-validation --write-q1-report`.

TABLE I
REAL HELD-OUT CIFAR-10 RECONSTRUCTION ACROSS FIVE RANDOM IMAGE SPLITS. LOWER MSE AND HIGHER PSNR/SSIM ARE BETTER.

Model	MSE	PSNR	SSIM
local-observables-only	$1.0024e-02 \pm 5.1e-03$	20.49 ± 2.12	0.7937 ± 0.0886
raw-linear-classical	$1.0024e-02 \pm 5.1e-03$	20.49 ± 2.12	0.7937 ± 0.0886
zz-only	$1.0763e-02 \pm 5.4e-03$	20.18 ± 2.11	0.7821 ± 0.0904
no-entanglement	$1.1194e-02 \pm 5.4e-03$	19.97 ± 2.01	0.7733 ± 0.0931
shuffled-pair-observables	$1.1651e-02 \pm 5.2e-03$	19.75 ± 1.92	0.7636 ± 0.0960
quadratic-classical	$1.1708e-02 \pm 5.6e-03$	19.76 ± 2.02	0.7653 ± 0.0945
HVK2D-real-cifar	$1.2197e-02 \pm 6.1e-03$	19.62 ± 2.06	0.7597 ± 0.0898
strict-classical-rff	$2.4169e-02 \pm 1.5e-02$	16.79 ± 2.20	0.5950 ± 0.1084
random-vqc	$8.7895e-02 \pm 5.7e-02$	11.29 ± 2.45	-0.0093 ± 0.1194

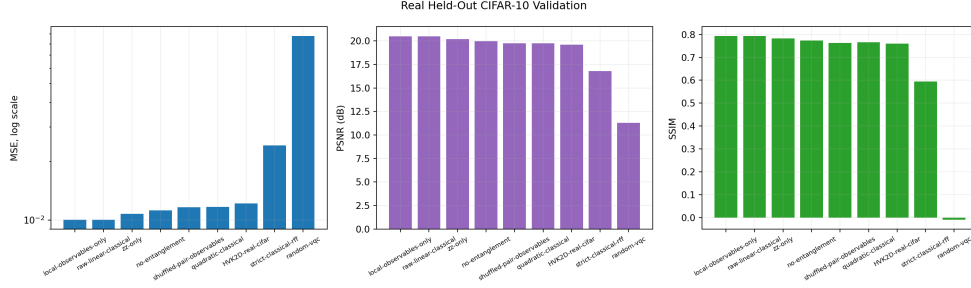


Fig. 1. Held-out CIFAR metric summary for HVK2D and same-width controls.

TABLE II
PAIRED STATISTICAL TESTS FOR HELD-OUT CIFAR RECONSTRUCTION.

Control	Pairs	Mean Δ PSNR	Bootstrap 95% CI	Wilcoxon p
raw-linear-classical	20	-0.87	$[-1.18, -0.62]$	1.91e-06
local-observables-only	20	-0.87	$[-1.18, -0.61]$	1.91e-06
no-entanglement	20	-0.35	$[-0.61, -0.10]$	0.0362
zz-only	20	-0.56	$[-0.78, -0.38]$	1.91e-06
quadratic-classical	20	-0.14	$[-0.36, 0.06]$	0.0826
strict-classical-rff	20	2.83	$[2.29, 3.43]$	1.91e-06
shuffled-pair-observables	20	-0.13	$[-0.45, 0.18]$	0.522
random-vqc	20	8.33	$[7.20, 9.52]$	1.91e-06

TABLE III
FINITE-SHOT OBSERVABLE-NOISE SIMULATION FOR REAL HELD-OUT CIFAR RECONSTRUCTION.

Shots	Mean MSE	Mean PSNR	Mean SSIM
128	$1.7964e-02$	17.83	0.6904
256	$1.7383e-02$	18.04	0.7067
512	$1.7106e-02$	18.15	0.7112
1024	$1.7388e-02$	18.13	0.7043
2048	$1.6882e-02$	18.21	0.7120
4096	$1.6929e-02$	18.22	0.7120
8192	$1.7016e-02$	18.21	0.7103

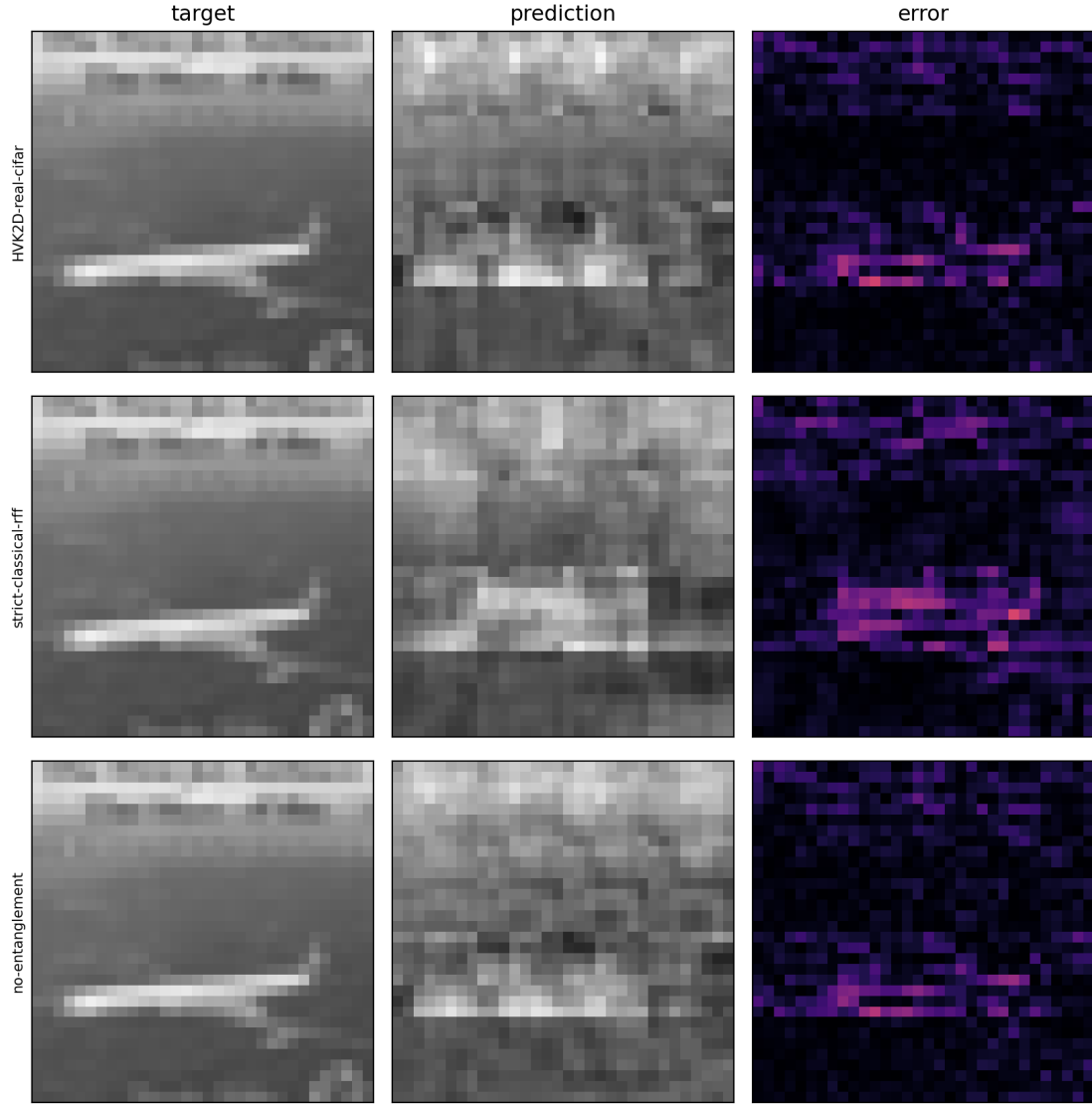


Fig. 2. Representative held-out CIFAR reconstruction panel. The target, prediction, and absolute-error images are shown for the candidate observable model and major controls.

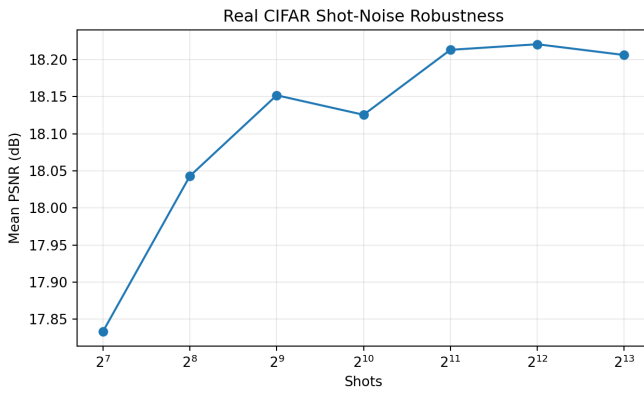


Fig. 3. Shot-noise robustness of the HVK2D observable feature vector on real held-out CIFAR splits.